

Project Design Phase Proposed Solution Template

Date	15 February 2025
Team ID	SWTID-2026-4968
Project Name	Script - controlled ACL - Restrict Record Access Based On Field Value
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	In many organizations, records such as incidents, HR cases and financial requests hold sensitive data, and standard role-based Access Control Lists (ACLs) can only allow or deny access at the table or role level. They cannot decide access based on the actual value stored in a record's field (for example, Department, Priority, Confidentiality or Assigned Group). As a result, users may see records they should not see, or administrators must create many roles and duplicate rules, which is hard to manage and prone to security gaps.
2.	Idea / Solution description	Implement a Script-controlled ACL that evaluates a server-side script at access time. The script reads the field value of the record being accessed (e.g., current.department, current.confidential) and compares it with the logged-in user's attributes (role, group, department). If the condition is satisfied, the script returns true and access is granted; otherwise it returns false and the record is restricted. The rule is applied on read, write, create and delete operations, so record access is controlled dynamically without creating extra roles.
3.	Novelty / Uniqueness	Access is decided dynamically at record level based on field values, instead of only static roles. A single reusable script can replace many role-based rules, giving fine-grained control with low maintenance. Conditions can be changed easily by editing the script, and the approach works together with existing role and condition checks for layered security.
4.	Social Impact / Customer Satisfaction	Protects sensitive personal and business data from unauthorized viewing, which increases user trust and helps meet privacy and compliance requirements. Employees see only the records relevant to them, giving a cleaner and safer experience. Administrators save time with simpler access management, and organizations reduce the risk of data leaks.

S.No.	Parameter	Description
5.	Business Model (Revenue Model)	Can be offered as a reusable security add-on or configuration package for enterprise platforms through a subscription or one-time licence fee. Additional revenue can come from implementation and customization services, security consulting, support and maintenance contracts, and training for administrators.
6.	Scalability of the Solution	The script-based rule can be reused across multiple tables, departments and applications by changing only the field and condition. It handles growing numbers of users and records without adding roles, and can be extended to new conditions such as location, business unit, or record status. Because the logic runs on the server with lightweight checks, it scales to large enterprise instances.